

Skills Summary

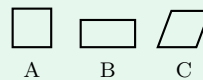
Sorting Four-Sided Shapes by What They Share

Use this reference while completing your worksheet or when studying.

Sorting by square corners

Rule: A **square corner** matches the corner of a sheet of paper. Slide the paper corner in: if it fits with no gap and no overlap, that corner is square. A four-sided shape with four square corners is a **rectangle**. Long sides never spoil a square corner.

Example: How many of these cards have four square corners?

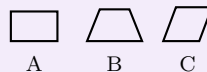


Step 1. The question is about corners only, so test corners and ignore how long the sides are.

Step 2. A fits four times; B fits four times; C leans, so two corners are too narrow and two too wide.

A and B fit, which is **2**

Try it: How many have four square corners?

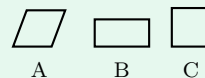


check: 1

Sorting by side lengths

Rule: Compare the four sides. All four the same length means the shape joins the “four equal sides” group: the **rhombus** and the **square**. A leaning shape can still have four equal sides.

Example: How many of these cards have four sides the same length?

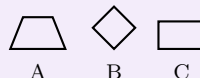


Step 1. The group is named by side length, so compare sides and leave the corners out of it.

Step 2. A leans but its four sides match; B has a long pair and a short pair; C is a square.

A and C fit, which is **2**

Try it: How many have four equal sides?

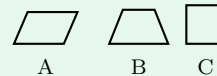


check: 1

Sorting by parallel sides

Rule: Two sides are **parallel** when they run the same way and would never meet, like train rails. Check both pairs of opposite sides: exactly one pair is a **trapezoid**, two pairs is a **parallelogram** (rectangles, rhombuses and squares are all in it).

Example: How many of these have two pairs of parallel sides?

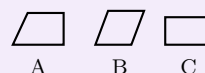


Step 1. Two pairs are asked for, so check both pairs on every card — one pair is not enough to stop.

Step 2. A: top and bottom run the same way, and so do the slanted sides. B: only the top and bottom do. C: both pairs do.

A and C fit, which is **2**

Try it: How many have *exactly one* pair of parallel sides?



check: 1

Reading a shape-sort graph

Rule: Each bar or tally row is one group, and its number tells how many shapes went in it. “How many in all” means add every group; “how many more” means subtract the smaller number from the larger one.

Example: A class sorted its cards: squares 7, trapezoids 4. How many more squares than trapezoids?

Step 1. “How many more” compares two groups, so this is a subtraction.

Step 2. Larger group 7, smaller group 4, so subtract: $7 - 4$.

There are **3** more squares

Try it: Same sort — squares 7, trapezoids 4. How many cards in all?

check: 11